

Advanced Portfolio Construction

Transforming Correlation into Distance

$$D(i,j) = \sqrt{0.5(1-\rho(i,j))}$$

$$D(i,j) = \sqrt{\sum_{k=1}^N (D(k,i) - D(k,j))^2}$$



Correlation
Distance



Systemic
Similarity



Asset
Clustering

Limitations of Traditional Portfolio Optimization

Traditional methods like Markowitz's mean-variance optimization (MVO) face significant challenges that limit their practical effectiveness:



Numerical Instability

Small changes in inputs can lead to drastically different portfolio allocations, creating unstable investment strategies.



Excessive Concentration

Traditional methods often result in portfolios concentrated in a small number of assets, reducing true diversification.



Poor Out-of-Sample Performance

Optimized portfolios often perform poorly when applied to data outside the sample used for optimization.

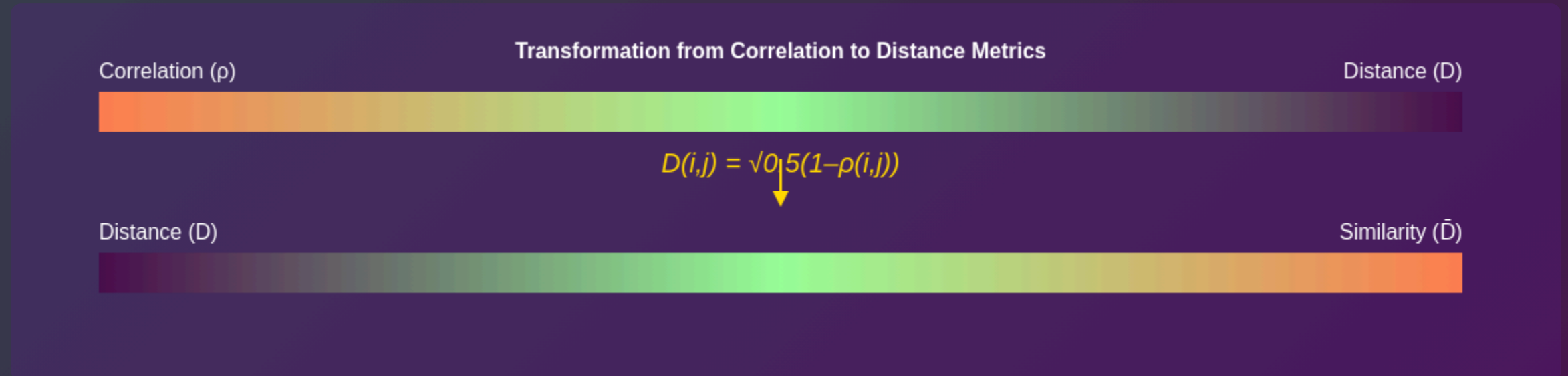
Root Cause



These issues stem from reliance on precise estimates of expected returns and covariance matrices, which are inherently noisy and prone to estimation errors.

Transforming Correlation into Distance Metrics

Traditional correlation measures are transformed into more informative distance metrics to effectively group assets based on their relationships.



Correlation-Distance Matrix (D)

Transforms correlation coefficients into a proper distance metric that satisfies non-negativity, identity of indiscernibles, symmetry, and triangle inequality properties.



Systemic Similarity Matrix ($\bar{D}$)

Measures how similarly assets are correlated with all other assets in the portfolio, providing insight into systemic relationships beyond pairwise correlations.



Purpose

These distance metrics enable more stable and intuitive portfolio construction, addressing the limitations of traditional methods by focusing on meaningful relationships between assets.

Correlation-Distance Matrix (D)

The correlation matrix is transformed into a correlation-distance matrix using the following formula:

$$D(i,j) = \sqrt{0.5 * (1 - \rho(i,j))}$$

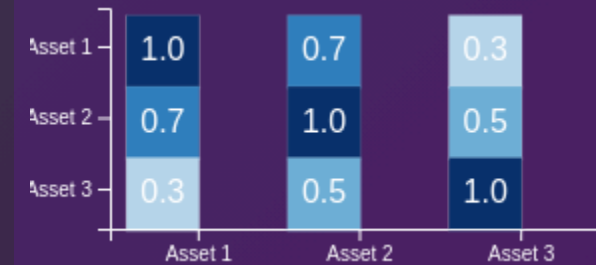
✓ Key Properties

- › Transforms correlation coefficients into distance metrics
- › Satisfies properties of true distance: non-negativity, identity of indiscernibles, symmetry, and triangle inequality

💡 Purpose

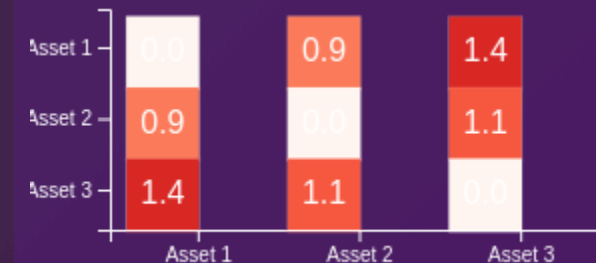
This transformation is essential for applying clustering algorithms, as they typically operate on distance measures rather than correlation values.

Correlation Matrix



➔ Apply Formula
 $D(i,j) = \sqrt{0.5 * (1 - \rho(i,j))}$

Correlation-Distance Matrix (D)



📌 Interpretation

$D(i,j)$ represents the direct dissimilarity between asset i and asset j , where higher values indicate greater dissimilarity.

Systemic Similarity Matrix ($\bar{D}$)

$$\bar{D}(i,j) = \sqrt{\sum_{k=1}^N (D(k,i) - D(k,j))^2}$$

The Systemic Similarity Matrix is formed by computing the Euclidean distance between all columns of the D matrix in a pair-wise manner.

💡 Interpretation

- ✅ **Lower value:** Indicates that assets i and j have similar correlation patterns with the rest of the portfolio
- ✅ **Higher value:** Indicates that assets i and j have different correlation patterns with the rest of the portfolio

🎯 Key Insight

$\bar{D}$ provides a measure of how similarly assets i and j are correlated with all other assets in the portfolio, capturing systemic relationships beyond simple pairwise correlations.

Example: $\bar{D}$ Matrix Calculation

D Matrix (Correlation-Distance)

1.0	0.3	0.7	0.8	0.3	1.0	0.6	0.9	0.7	0.6	1.0	0.5	0.8	0.9	0.5	1.0
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$\bar{D}$ Matrix (Systemic Similarity)

0.0	0.5	0.7	0.8	0.5	0.0	0.4	0.6	0.7	0.4	0.0	0.5	0.8	0.6	0.5	0.0
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Understanding D vs. D-bar



Correlation-Distance Matrix (D)

$$D(i,j) = \sqrt{0.5(1-\rho(i,j))}$$

Direct distance between assets i and j, transforming correlation into a true distance metric.



Systemic Similarity Matrix ($\bar{D}$)

$$D(i,j) = \sqrt{\sum_{k=1}^N (D(k,i) - D(k,j))^2}$$

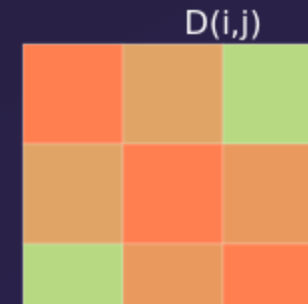
Measures how similar assets i and j are correlated with all other assets in the portfolio.



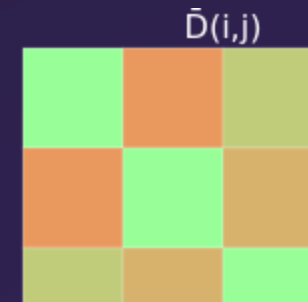
Key insight: Lower $\bar{D}$ values indicate higher systemic similarity between assets.

Visual Comparison

D Matrix: Direct distance between assets



$\bar{D}$ Matrix: Systemic similarity with other assets



■ Lower value ■ Higher value

Hierarchical Asset Clustering Process

Hierarchical clustering groups assets based on distance metrics, revealing deeper relationships beyond simple correlations.

Recursive, Bottom-Up Approach



Starts with each asset as its own cluster, merging closest pairs iteratively.

Linkage Method



Determines distance between clusters, guiding which clusters to combine.

Dendrogram Structure



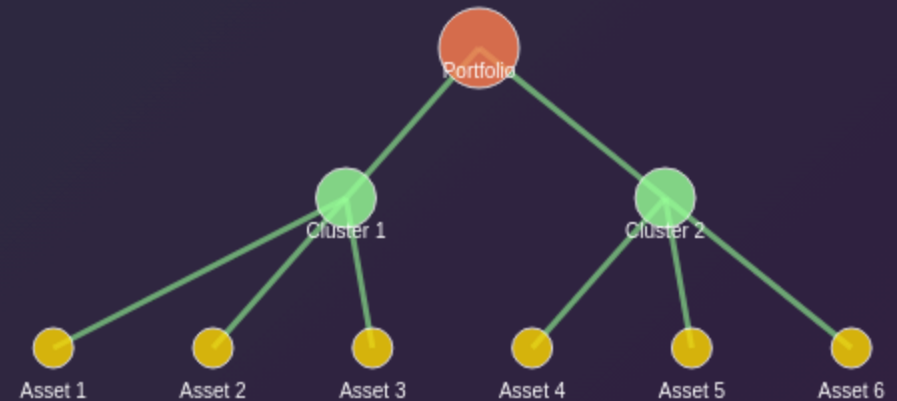
Visual representation of hierarchical relationships between assets.

Key Benefits



Addresses limitations of traditional methods by focusing on systemic relationships rather than isolated correlations.

Hierarchical Clustering Visualization



Each merge represents combining assets with similar correlation patterns

Forming the First Cluster

Identifying the First Cluster

$$U[1] = \operatorname{argmin}_{(i,j)} D(i,j)$$

The first cluster is formed by identifying the pair of assets with the minimum value in the $\bar{D}$ matrix, which represents the highest systemic similarity.

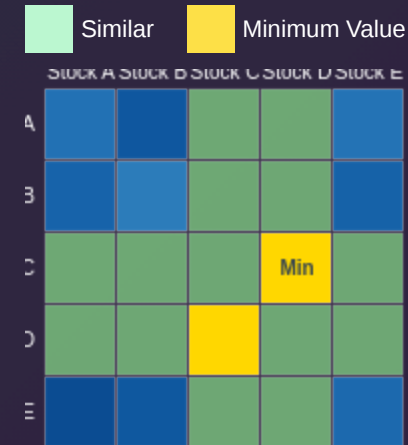
1 Find Minimum Value

Identify the minimum value in the $\bar{D}$ matrix, which represents the closest pair of assets based on their correlation profiles.

2 Form First Cluster

Combine the assets with the minimum $\bar{D}$ value into the first cluster, as they have the most similar correlation patterns with the rest of the portfolio.

Visualizing the First Cluster Formation



$$i + j = U[1]$$

Applications and Implications

Practical Applications

Hierarchical Risk Parity (HRP)

Creates robust investment portfolios with diversified risk contributions across asset classes.

Portfolio Construction

Addresses numerical instability and over-concentration issues in traditional optimization.

Systemic Risk Management

Groups assets by their systemic properties rather than simple correlations.

Key Implications

This clustering methodology establishes a more stable foundation for portfolio diversification by:

- Reducing numerical instability
- Preventing excessive concentration
- Creating intuitive portfolio structures

★ Conclusion

This approach transforms correlation into meaningful distance metrics, enabling a more robust and intuitive foundation for diversified investment strategies.